Rental Price Prediction Using Multiple Linear Regression

Problem Statement

1. Introduction

The real estate rental market is growing rapidly due to urbanization and increasing demand for residential properties. Determining the correct rental price is an important task for property owners, tenants, and real estate agencies. Rental prices depend on various factors such as property area, number of bedrooms, bathrooms, age of the property, and location. Traditional methods of estimating rent rely on experience and market comparisons, which are often time-consuming and inaccurate.

Machine Learning provides an effective solution by analyzing historical rental data and predicting rental prices automatically. **Multiple Linear Regression (MLR)** is a supervised learning algorithm that predicts rental prices based on multiple property features. This project develops a machine learning model that estimates rental prices accurately, helping users make better rental decisions while reducing manual effort and pricing errors.

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2. Problem Statement

Estimating the rental price of a property manually is difficult because many factors influence the final rent. Property owners often rely on personal experience or local market comparisons, which may result in incorrect pricing. Overpricing can reduce tenant interest, while underpricing leads to financial loss.

The proposed system uses **Multiple Linear Regression** to analyze historical rental property data and predict rental prices automatically. The system provides quick, accurate, and reliable rental estimates based on property features, improving transparency and efficiency in the rental market.

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3. Problem Definition

The main problem is the absence of an accurate and automated system for rental price estimation. Manual pricing methods are time-consuming, inconsistent, and prone to human error. The project aims to develop a machine learning-based prediction system that uses Multiple Linear Regression to estimate rental prices from property attributes such as area, bedrooms, bathrooms, and age. The system helps landlords, tenants, and real estate agencies make informed decisions with reliable rental price predictions.

4. Features

1. **Automatic Rental Price Prediction** based on property details using Multiple Linear Regression.
2. **User-Friendly Interface** for entering property information and viewing predicted rent.
3. **Historical Data Analysis** to identify relationships between property features and rental prices.
4. **Multiple Input Parameters** including area, bedrooms, bathrooms, and property age.
5. **Fast and Accurate Predictions** using a trained machine learning model.
6. **Model Performance Evaluation** using MAE, MSE, RMSE, and R^2 Score.
7. **Scalable System Design** that supports future enhancements such as location, parking, furnishing status, and advanced algorithms.

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5. Advantages

- Provides accurate and consistent rental price estimation.
- Reduces manual effort and calculation time.
- Minimizes pricing errors and human bias.
- Helps landlords determine fair rental prices.
- Assists tenants in comparing rental costs.
- Improves efficiency for real estate agencies.
- Supports data-driven decision-making.
- Easy to use with a simple graphical interface.
- Can be enhanced with additional property features.
- Cost-effective and reliable for small and medium-scale applications.

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6. Conclusion

The **Rental Price Prediction Using Multiple Linear Regression** project presents an efficient and intelligent solution for estimating property rental prices using machine learning techniques. By analyzing historical rental data and considering multiple property attributes, the system provides fast, accurate, and reliable rental price predictions. It eliminates the drawbacks of traditional manual estimation methods, such as inconsistency, human error, and excessive time consumption.

The project benefits landlords, tenants, and real estate agencies by providing fair and data-driven rental price estimates. It also demonstrates the practical application of Multiple Linear Regression in solving real-world prediction problems. With future enhancements such as location-based analysis, real-time property data, and advanced machine learning models, the system can become even more accurate and useful. Overall, this project contributes to improving transparency, efficiency, and decision-making in the real estate rental market.